

Henry Ye

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TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java, Rust, Golang, TypeScript, JavaScript, HTML, CSS, SQL, ABAP
- **Frameworks & Libraries:** Spring Boot, Spring Cloud, PostgreSQL, Redis, React, React Native, Angular, Next.js, Express, Mongoose, Flask, SQLite
- **Tools & Platforms:** Docker, Kubernetes, Git (GitHub, GitLab), MongoDB, PostgreSQL, Elasticsearch, RabbitMQ, RocketMQ, Postman, JMeter, AWS (ECR, ECS, CloudFormation, CodePipeline, DynamoDB), Travis CI, Heroku, Vercel, SAPUI5, SAP HANA, Trello, Figma

EDUCATION

- **University of Toronto** Toronto, Ontario
Master of Engineering - Electronic Computing Engineering; GPA: 3.97/4 Sep 2023 - Dec 2024
- **Simon Fraser University** Burnaby, British Columbia
Bachelor of Science - Computer Science; GPA: 3.59/4.33 May 2019 - Dec 2022

TECHNICAL WORK EXPERIENCE

Software Developer

Shanghai, China

SAP Full-time Co-op

Oct 2021 - Mar 2022

- Implemented a modern front-end using **JavaScript**, **SAPUI5**, and **Fiori UI**, replicating the legacy interface within 1.5 weeks and improving code maintainability while increasing unit test coverage from (50% to 80%).
- Refactored and optimized Core Data Services (**CDS**) schemas in **ABAP**, streamlining queries with fewer table joins from 3 to 2, improving db query performance, reducing the count of temporary tables to reduce storage costs, and collaborating cross-functionally with multiple teams to redesign and streamline database schemas.
- Implemented an event-driven architecture with **RocketMQ** to enable decoupled, asynchronous transaction management and efficiently handle burst traffic through asynchronous job processing.

TECHNICAL PROJECTS

Distributing Online Shopping System

University of Toronto

June 2025 - Aug 2025

- Developed and optimized online shopping **CRUD** operations using **Spring Boot** with **MyBatis**, leveraging **MySQL** for persistent storage and **AWS DynamoDB** for distributed system scalability.
- Implemented **distributed lock** with **Redis** and **Lua** scripts using the Try-Confirm-Cancel (TCC) pattern to ensure inventory consistency and prevent overselling, while enhancing the Prime Day Sale module with a Redis-based purchase limit feature that reduced MySQL traffic load by 90% during peak order bursts.
- Enhanced product search performance by integrating **Elasticsearch** inverted indexing, achieving a 90% improvement in query speed compared to MySQL fuzzy search.
- Designed and executed a Prime Day Sale event system capable of sustaining 10K QPS burst traffic under **JMeter** stress testing.
- Deployed and configured **Spring Cloud Consul** clusters to enable service discovery and registration across distributed microservices.
- Containerized services with **Docker** and managed images via **AWS ECR**; expanded **CodePipeline** and **CodeBuild** to establish automated CI/CD workflows on **AWS ECS**, accelerating deployment cycles and improving rollout efficiency by 60%.

Rental Property Web Application

University of Toronto

May 2025 - June 2025

- Developed a full-stack rental platform using **Next.js**, **JavaScript**, **HTML**, and **TailwindCSS**, allowing users to list, search, and manage property listings and deployed the website to **Vercel**.
- Implemented authentication via **Google OAuth** and **NextAuth.js** to enable user registration and authentication.
- Designed and developed **RESTful APIs** to handle property listings, user interactions, and authentication, ensuring scalable and maintainable client-server communication.
- Designed and integrated property listing **CRUD** functionalities backed by **MongoDB** and **Mongoose**.
- Integrated multi-image uploads with **Cloudinary** and implemented dynamic map views using **Mapbox**.